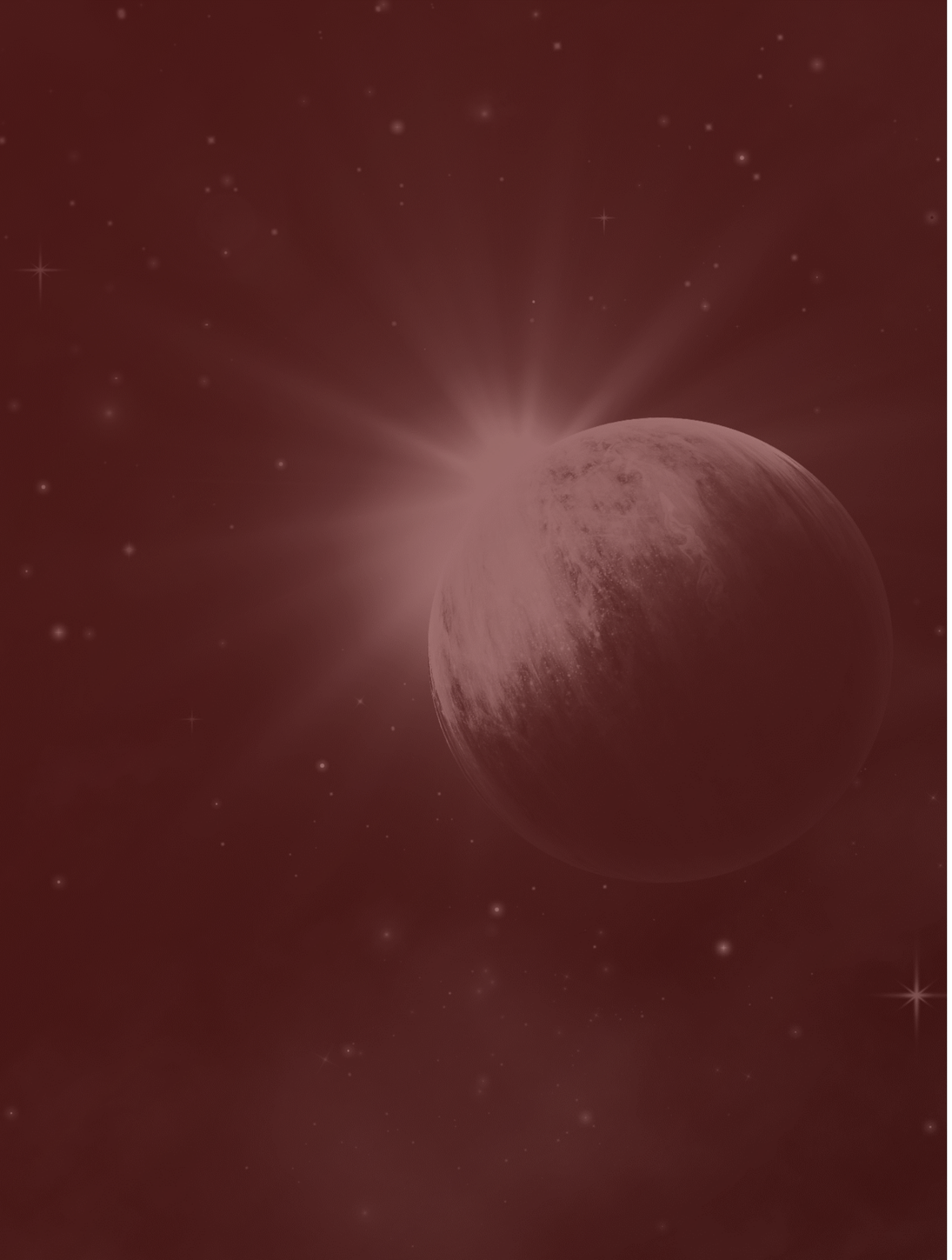
Evolute Migrate

AI-based Containerization

Kristopher Francisco

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United States

www.evolute.io | @evoluteIO



Agenda

- Overview
- Evolute Migrate
  - BCD Framework
  - Demo
- Scalability Patterns

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Overview

History

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Timeline

First and Most Comprehensive



2016
Evolute introduces first containerization*

2017
Google introduces first-generation of its containerization software

2018
Evolute develops data-science/AI framework for modernization (1st Generation Cloud Digital Transformation)

Google acquires Velostrada Forms Google Migrate

2019
Docker releases MTA program

2020
Amazon introduces App2Container

Evolute introduces Containerization AI (2nd Generation Containerization)

2021
Microsoft introduces Azure Migrate (containerization)

2022
Konveyor OSS VM + Containerization (Image Repository) introduced

* USPTO 62797200

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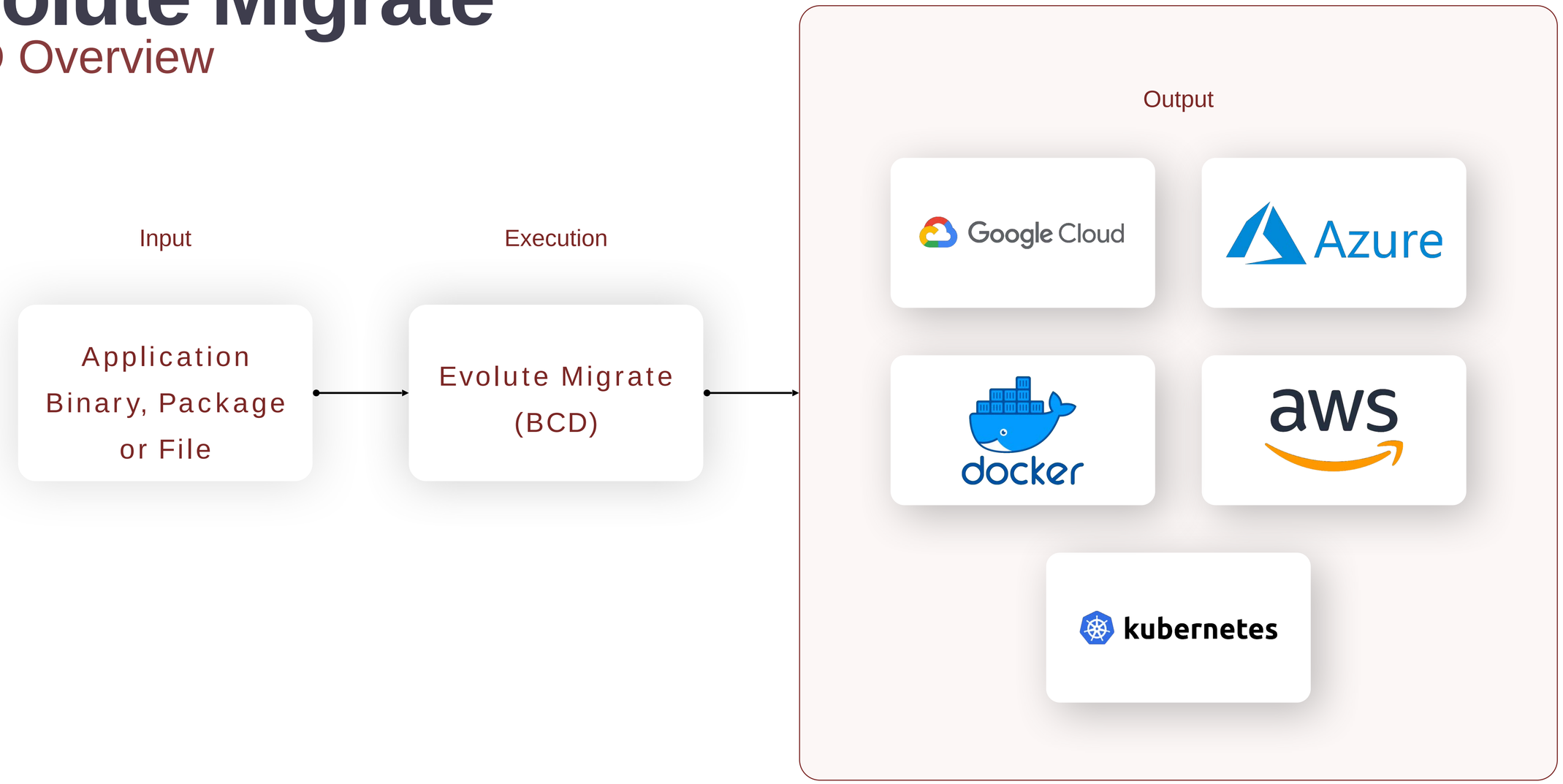
Evolute Migrate

BCD Overview

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Evolute Migrate

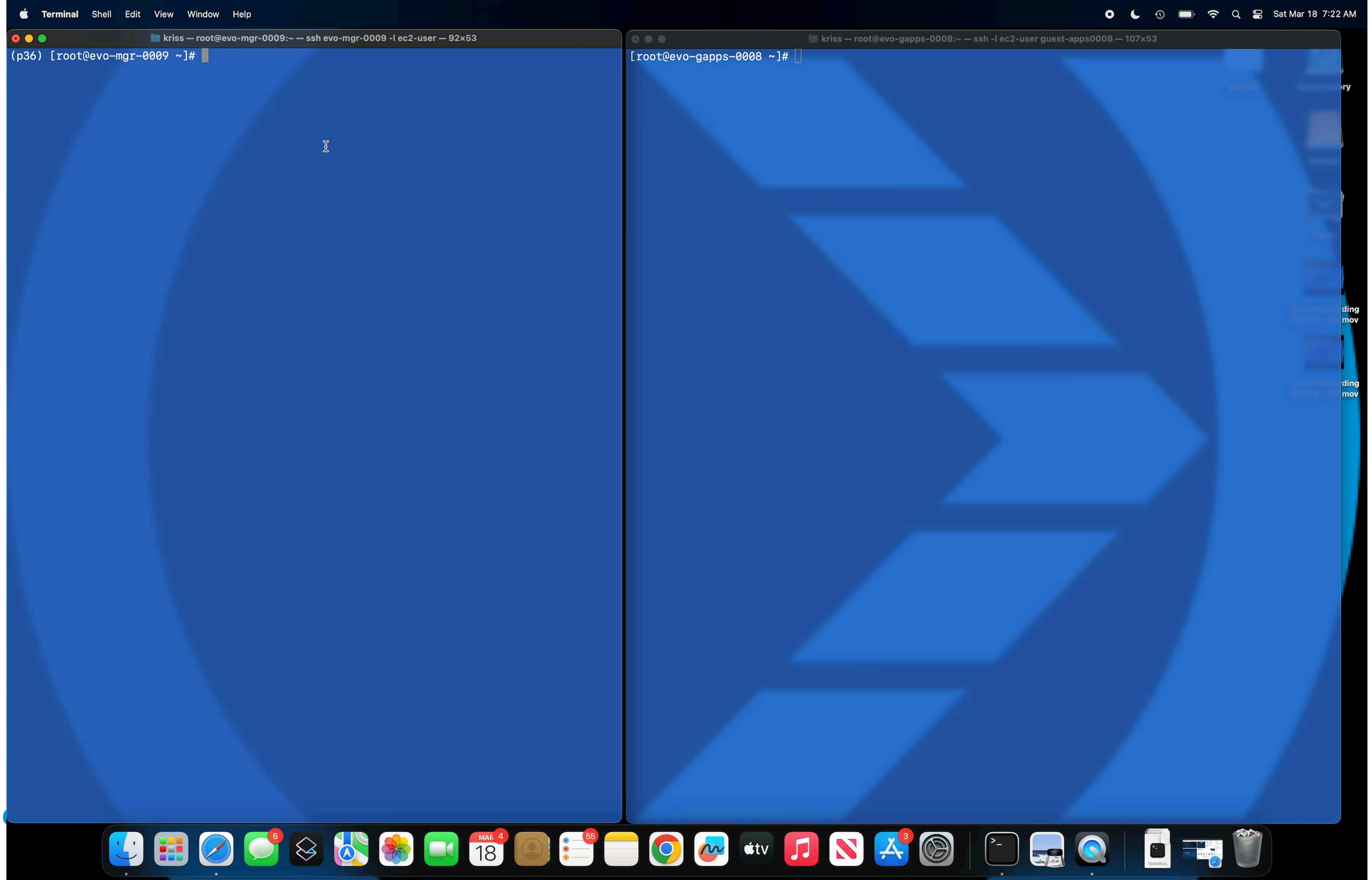
BCD Overview



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Demo

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Terminal Shell Edit View Window Help
Sat Mar 18 7:22 AM
kriss — root@evo-mgr-0009:~ — ssh evo-mgr-0009 -l ec2-user — 92×53
(p36) [root@evo-mgr-0009 ~]#
kriss — root@evo-gapps-0008:~ — ssh -l ec2-user guest-apps0008 — 107×53
[root@evo-gapps-0008 ~]#

Evolute Migrate

BCD Deep Dive

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AI Framework

BCD

Classification

Leverages K-Means Clustering via Unsupervised ML

Containerization

- Leverages Decision Trees via Supervised ML
- ~3100 Base Paths (OS, Classification Input, Output Artifact, Destination Platform)

Runtime

- Run In Any Target Docker/K8S Platform
- Start Services
- Separate Components

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Evolute Migrate Model

BCD - Binary Configuration Data

CLASSIFICATION

Binary, Package or File

Example binary - mysqld

Example package - ftp-0.17-67.el7.x86_64.rpm

Example binaries(nginx, mssql, customer vendor software)

PROFILE

CONFIGURATION

Binary Configuration Data Mapping

Software Language Evaluation

Library and Dependencies Mapping

OS Source and Destination Mapping

Software Static and Dynamic Variable Analysis

Storage and Data Transformation

TRANSFORM

CONTAINERIZE

Properly Instantiate

Separate Services

Start Order

Proper Clustering

Scalability

RUN

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Evolute Migrate

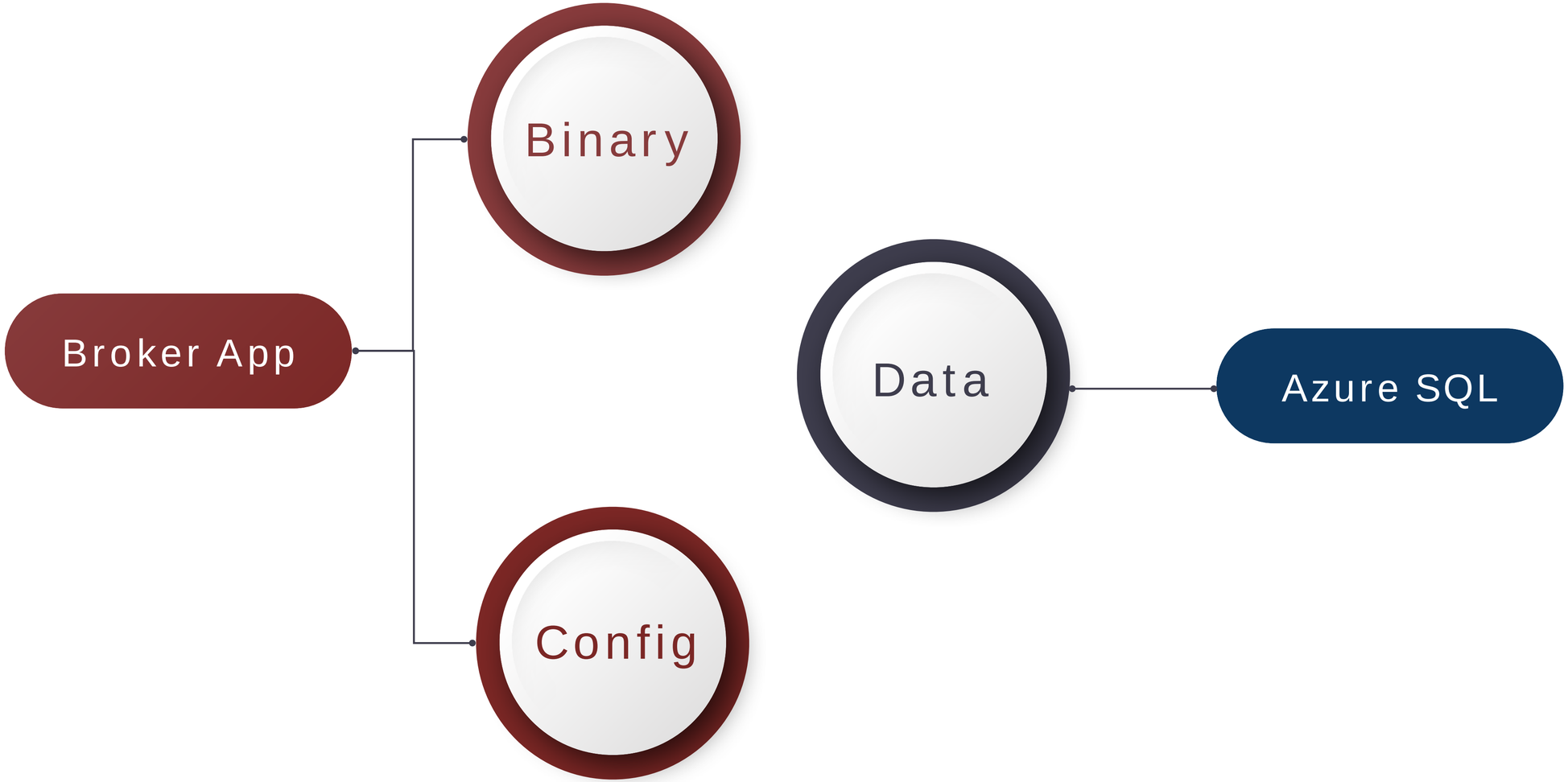
Example App

In-House Developed Software

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Evolute Migrate

Example Software



Note: Not all components are depicted (e.g. client side software, serverless API, et al)

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Evolute Migrate

Software Manifest

Broker App

Binary - Software Runtime

Configuration - Unique Variables Make Instance Unique

```
FROM mcr.microsoft.com/windows/servercore:ltsc2022
SHELL ["cmd", "/c"]


# Runtime dependencies
ADD ["https://download.microsoft.com/download/9/3/F/93FCF1E7-E6A4-478B-96E7-D4B285925B00/vc_redist.x64.exe" , "C:\\\\msvcrt-2015-x64.exe"]
RUN C:\msvcrt-2015-x64.exe /quiet /install /wait && del C:\msvcrt-2015-x64.exe || exit 0
ADD ["https://download.microsoft.com/download/9/3/F/93FCF1E7-E6A4-478B-96E7-D4B285925B00/vc_redist.x86.exe" , "C:\\\\msvcrt-2015-x86.exe"]
RUN C:\msvcrt-2015-x86.exe /quiet /install /wait && del C:\msvcrt-2015-x86.exe || exit 0

# Application: Broker Services
RUN md "C:\\Program Files (x86)\\Custom Company" || exit 0
COPY ["C\\\\Program Files (x86)\\\\Custom Company\\\\Broker Services", "C:\\\\Program Files (x86)\\\\Custom Company\\\\Broker Services"]
RUN sc create "BrokerService" binpath="\"C:\\Program Files (x86)\\Custom Company\\Broker Services\\Custom Company.OutlookIntegration.BrokerService.exe\"" start=auto DisplayName="Broker Broker Service"

# Setup/Create MSMQ Queues
SHELL ["powershell", "-command"]
RUN Set-Item Env:PSModulePath -Value 'C:\Windows\system32\WindowsPowerShell\v1.0\Modules;C:\Program Files\WindowsPowerShell\Modules;C:\Users\ContainerAdministrator\Documents\WindowsPowerShell\Modules'

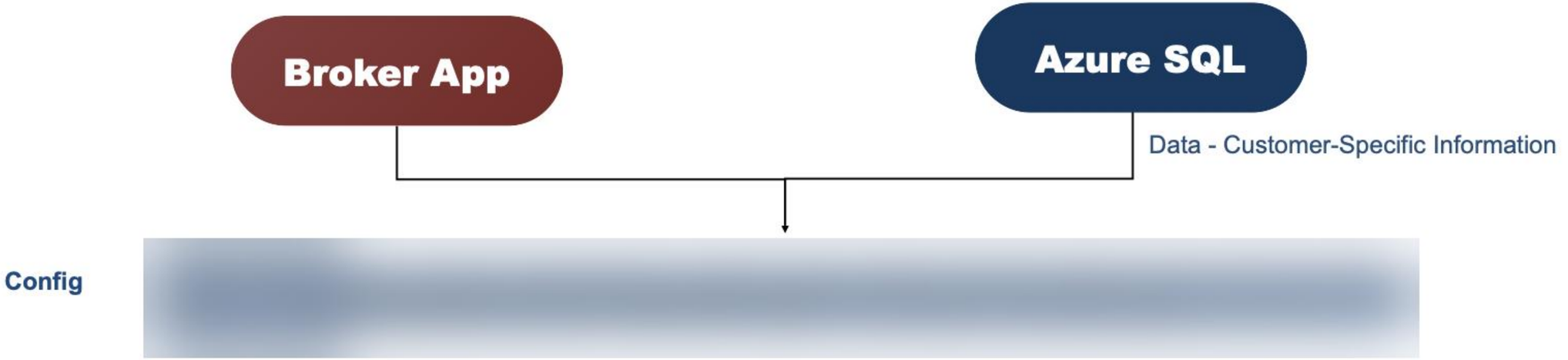
RUN Install-WindowsFeature MSMQ-Services
RUN New-MsmqQueue -Name "broker" -Label "broker" -QueueType Private -Transactional | Set-MsmqQueueAcl -UserName "Everyone" -Allow FullControl
RUN New-MsmqQueue -Name "booking" -Label "booking" -QueueType Private -Transactional | Set-MsmqQueueAcl -UserName "Everyone" -Allow FullControl
RUN New-MsmqQueue -Name "messageprocessor" -Label "messageprocessor" -QueueType Private -Transactional | Set-MsmqQueueAcl -UserName "Everyone" -Allow FullControl


#Create Services
SHELL ["cmd", "/c"]
RUN sc create "DataPullService" binpath="\"C:\\Program Files (x86)\\Custom Company\\Broker Services\\Custom Company.OutlookIntegration.DataPullService.exe\"" start=auto DisplayName="Broker Data Pull Service"
RUN sc create "MessageProcessorService" binpath="\"C:\\Program Files (x86)\\Custom Company\\Broker Services\\Custom Company.OutlookIntegration.MessageProcessorService.exe\"" start=auto DisplayName="Broker Message
Processor Service"
RUN sc create "DataWriterService" binpath="\"C:\\Program Files (x86)\\Custom Company\\Broker Services\\Custom Company.OutlookIntegration.DataWriterService.exe\"" start=auto DisplayName="Broker Data Writer Service"
```

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Evolute Migrate

App Config (Example)



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Architecture + Scalability

Scalability Patterns

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Scalability Patterns

Deterministic Architectures



By Application Function

ex: Shopping Cart function on e-commerce app



By Resource Demand

ex: component requiring more hardware/network resources



By Component

ex: nginx package installed on OS

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Scalability Patterns

Deterministic Architectures



By Application Function

Scalability Pattern: **m x n**

Implementation: e.g., **Deployments + Services**



By Resource Demand

Scalability Pattern: **n+1**

Implementation: e.g. **Pod Autoscaler**



By Component

Scalability Pattern: **n+1**

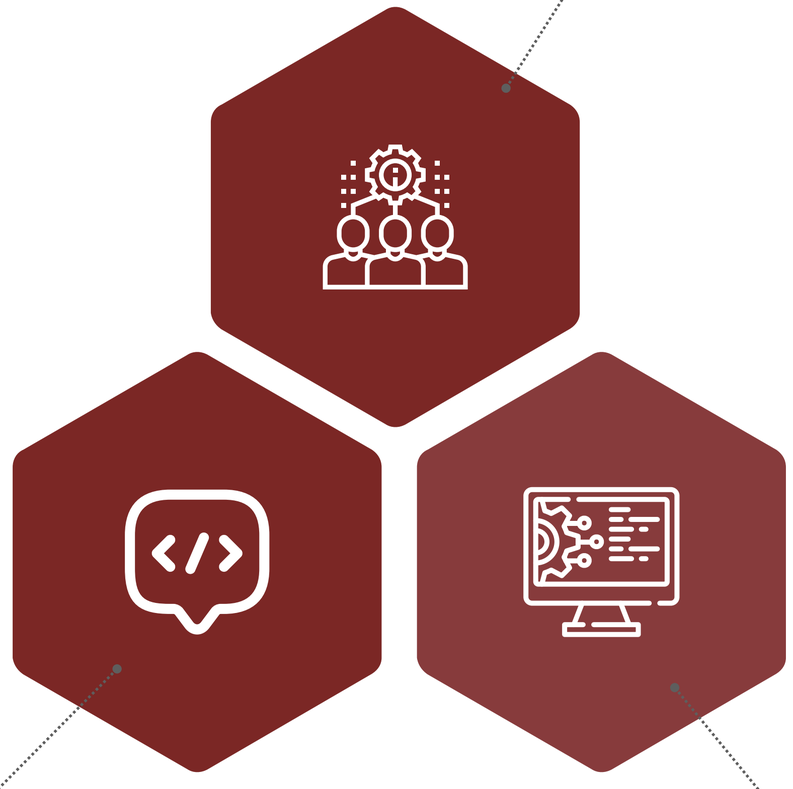
Implementation: e.g., **Deployment Count**

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Scalability Patterns

A Few Computational Truths

Everyone respects OSI



Communication
expresses Relationships

Operating Systems
are abstractions

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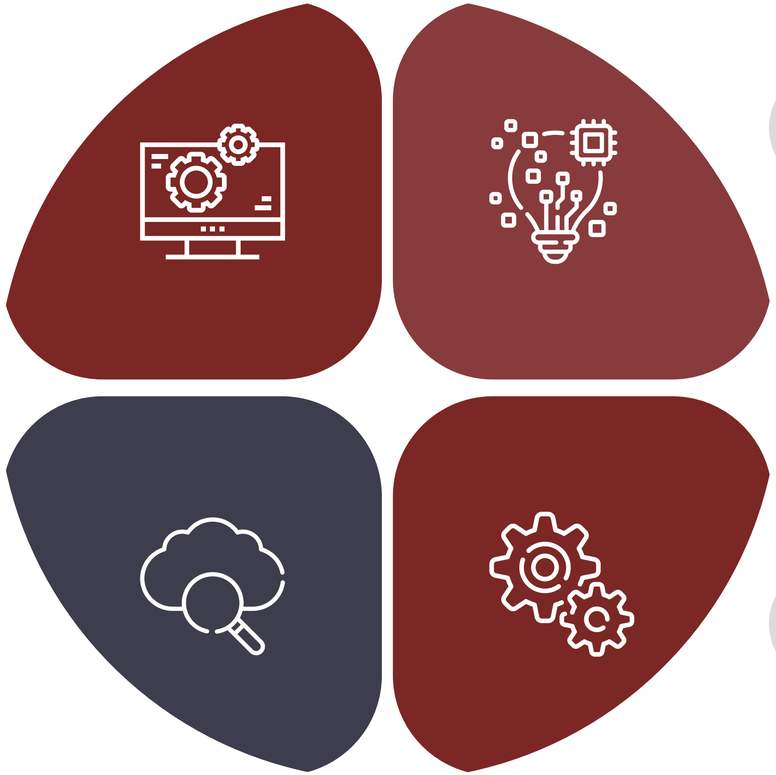
Scalability Patterns

Use Cases

Egde Computing

MEC

Distributed Workloads



Enterprise Digital Transformation

Software Modernization

OS Modernization

Hybrid/Multi Cloud

Public Cloud (AWS, Azure, GCP)

Private Cloud

Function Driven Environments

Serverless/Lambda

No Code/Low Code

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Thank You

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